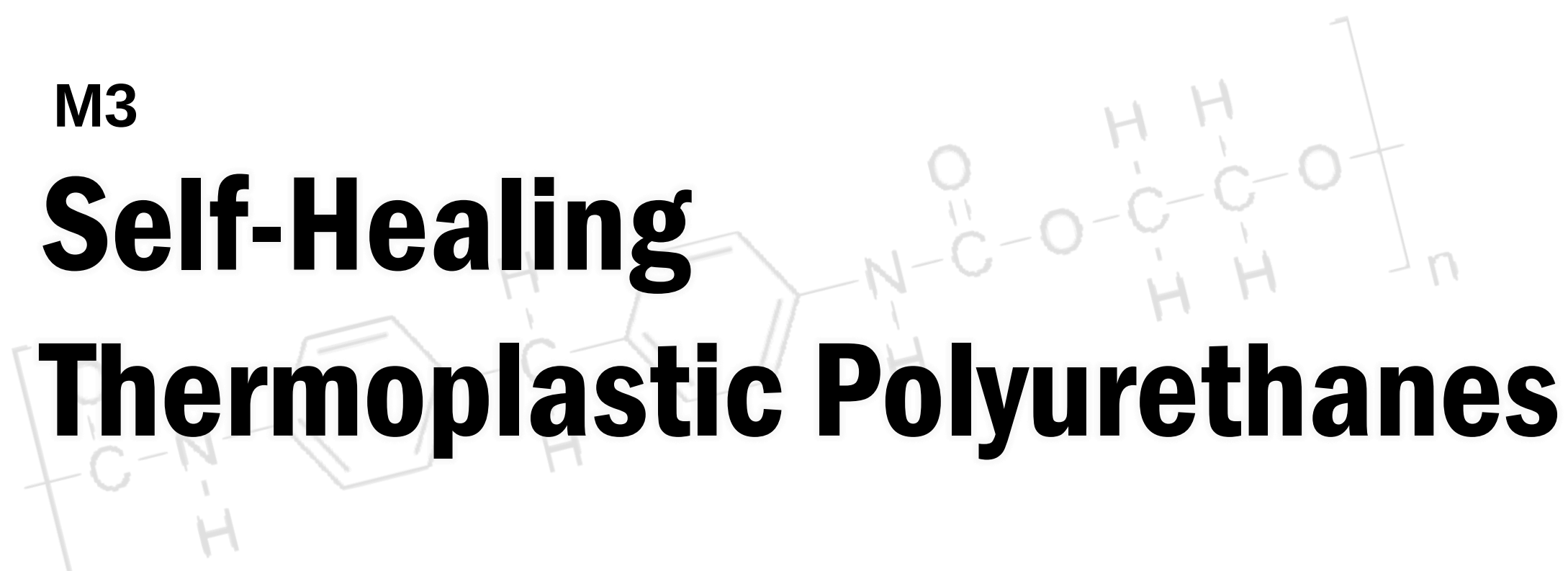


M3

Self-Healing Thermoplastic Polyurethanes



Yuxin Liu



Jingyu Chen



Zishan Ding



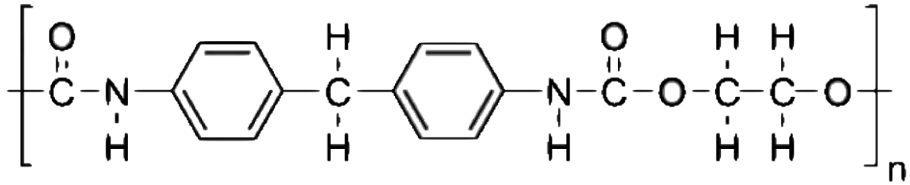
Tianjian Feng



Zeying Jin



Guozhi Xu



Content



Introduction



Self-healing Mechanism



Mechanical Properties



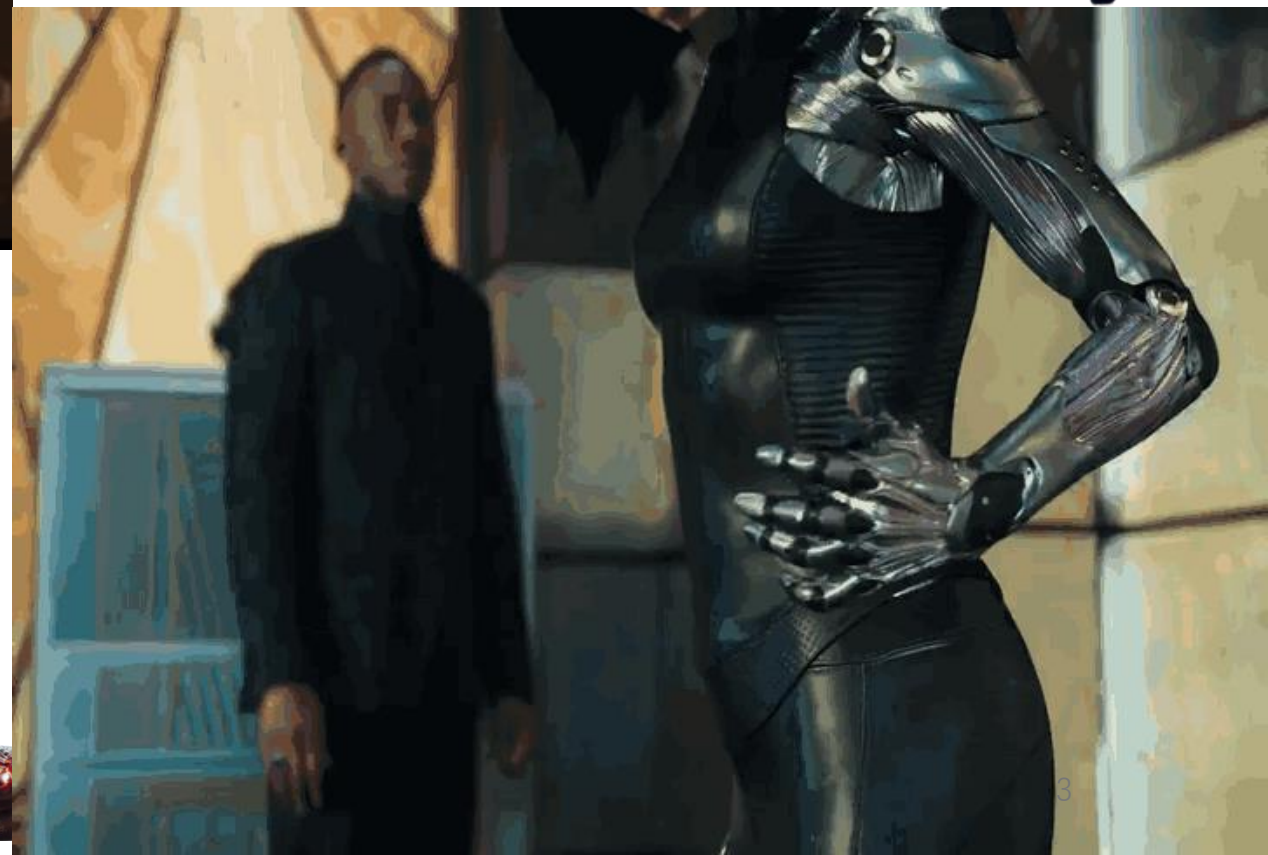
Application & Sustainability



Alita: Battle Angel
Alita: Cybernetic Body



Avengers 3: Infinity War
Iron Man: Mark50



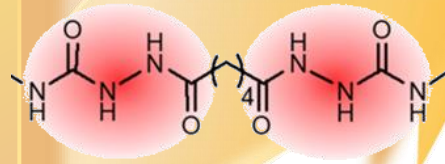


0 |

Introduction

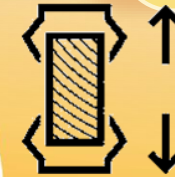
Have some modifiable active sites

Modifiable to form S-H material



Good Mechanical properties

- Suitable strength
- Good Elasticity
- High toughness



How To Choose?

High sustainability



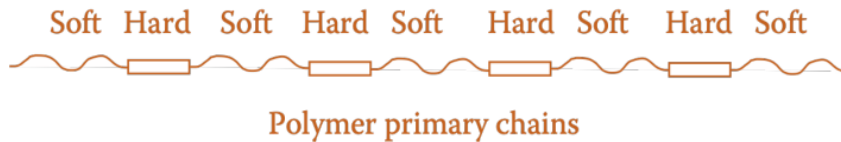
High processability

Introduction

TPU

1. Thermoplastic Elastomer

Soft segment + Hard segment



2. Low Tg: $-71^{\circ}\text{C} \sim -32^{\circ}\text{C}$ ^[1]

Rubbery at room temperature
→ Self-healing at room temperature

3. Low crystallinity & cross-link density

High elasticity and toughness
→ Withstand damage and recovery in self-healing

4. Modifiable site ^[2]

The benzene ring → the introduction of disulfide
Multiple hydrogen bond system

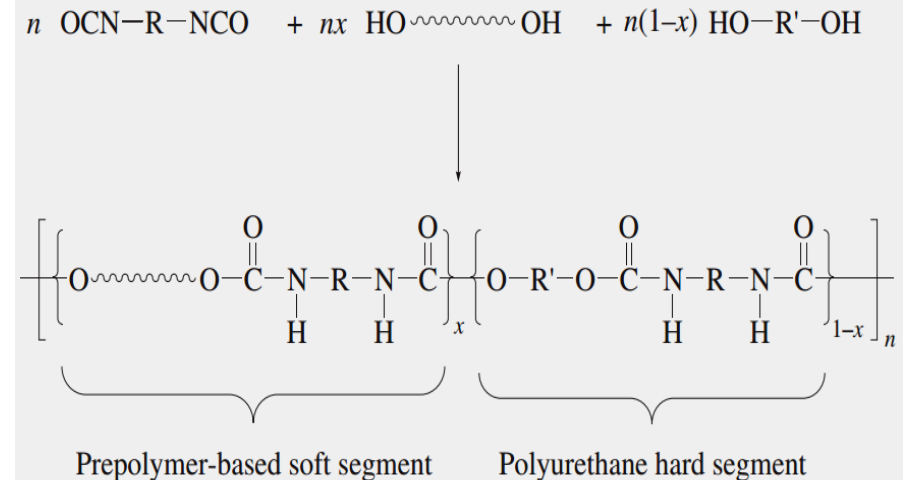
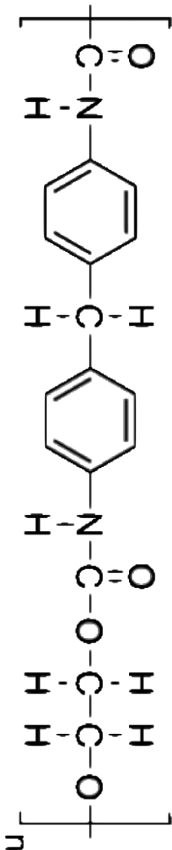


Fig.1 The synthesis of the TPU^[3]

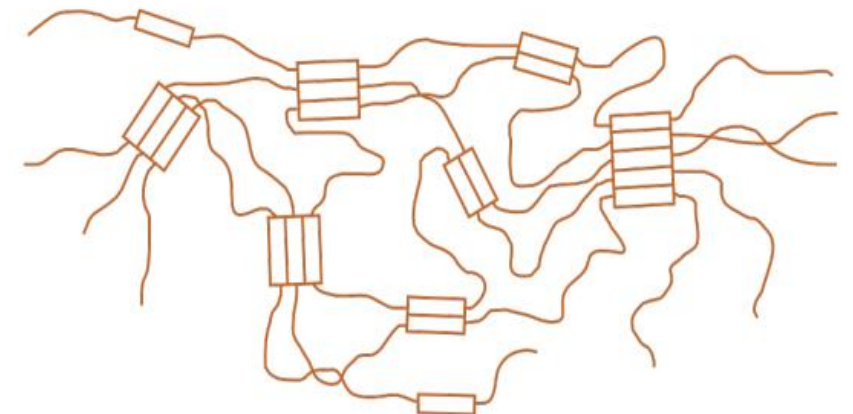


Fig.2 The schematic of the TPU structure ^[3]

02



Self-healing Mechanism



02

Self-healing Mechanism

● Molecular structure

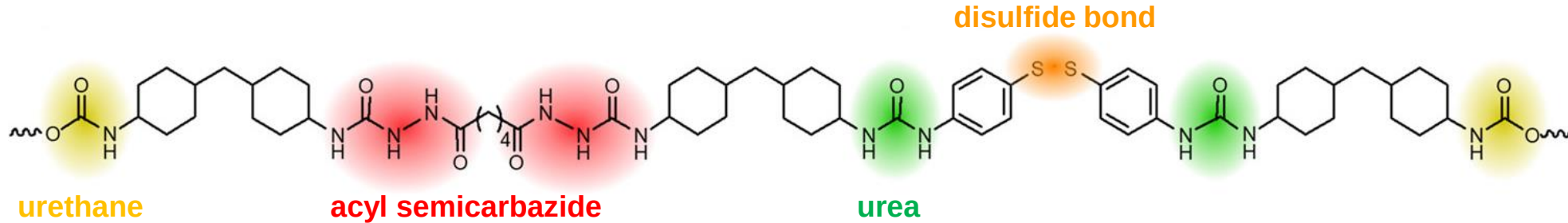


Fig. 3 Molecular structure of self-healing TPU with multiple dynamic bonds [4].

● Self-healing mechanism 1: noncovalent interaction

Multiple hydrogen bond system

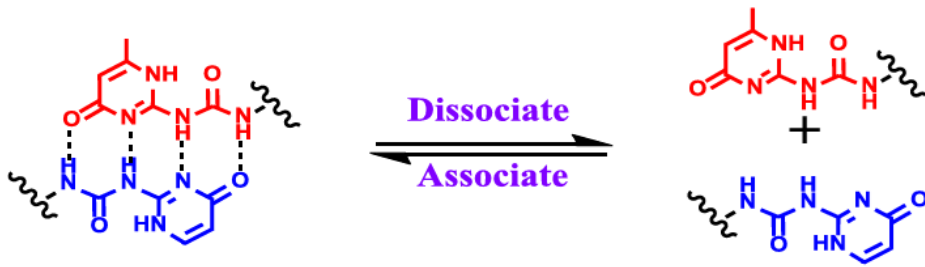


Fig. 4 Self-healing mechanism of multiple hydrogen bonds [5].

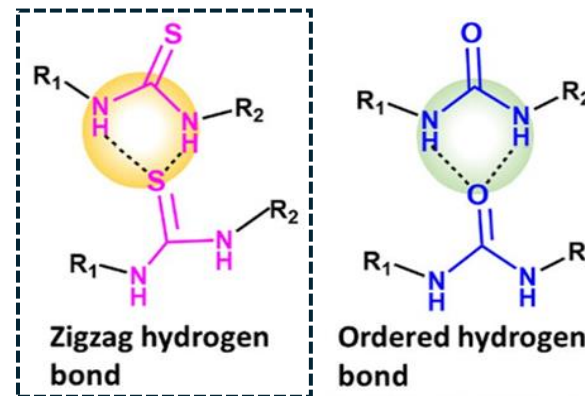


Fig. 5 Types of hydrogen bonds [5].

Zigzag H bond preferred:

- Reduce crystallization compared with ordered H bond
- Better mobility of molecular chain
- Higher efficiency of self-healing

The dynamic nature of hydrogen bonding facilitates rapid healing.



02

Self-healing Mechanism

● Molecular structure

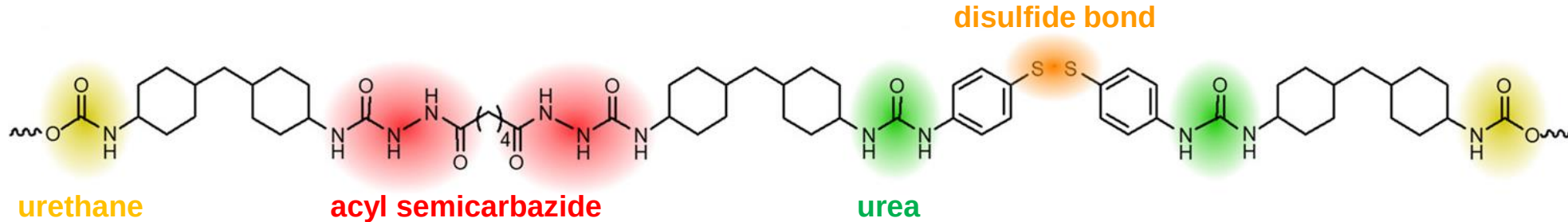


Fig. 3 Molecular structure of self-healing TPU with multiple dynamic bonds [4].

● Self-healing mechanism 2: covalent interaction

Disulfide bond

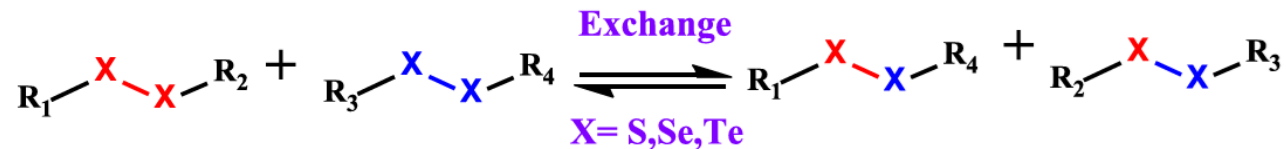


Fig. 6 Self-healing mechanism of dynamic covalent bonds [5].

● Synergistic effect^[4]

- Hydrogen bonds: provide rapid, low-energy healing
- Disulfide bonds: provide durable long-term structural backbone reformation

*Synergistic effect enables **fast** and **reliable** healing process.*

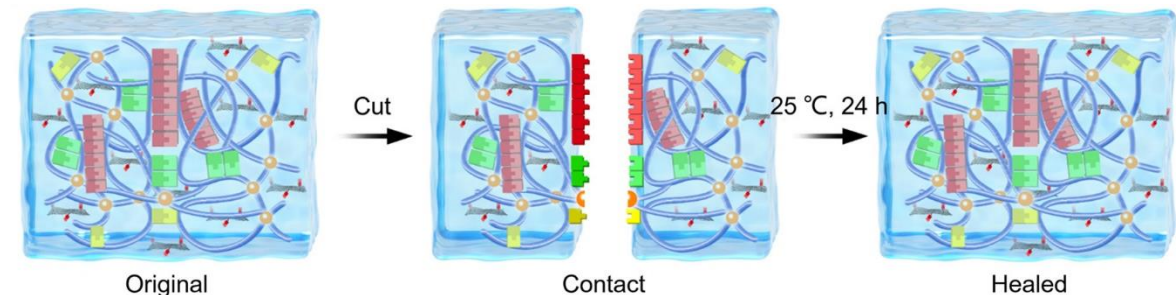


Fig. 7 Schematic diagram of TPU self-healing mechanism [4].

03



Mechanical Properties



03

Self-healing Efficiency

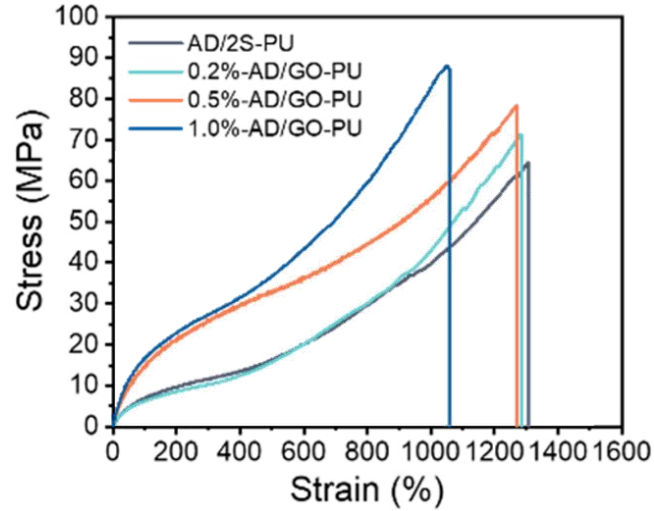


Fig.8 σ - ϵ curves of TPUs before and after healing(with additive) [6]

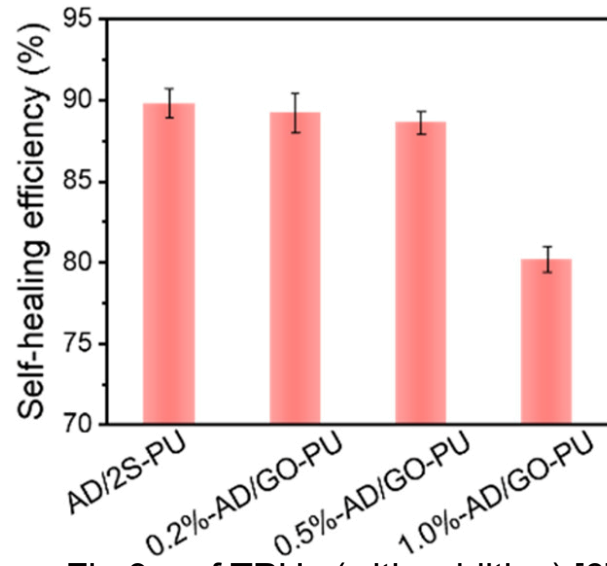
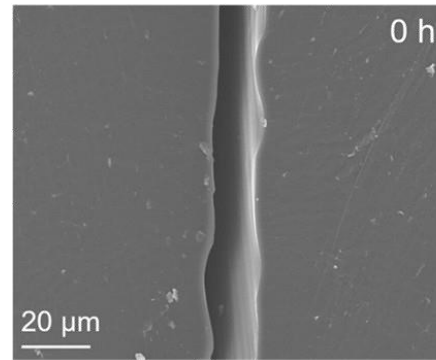
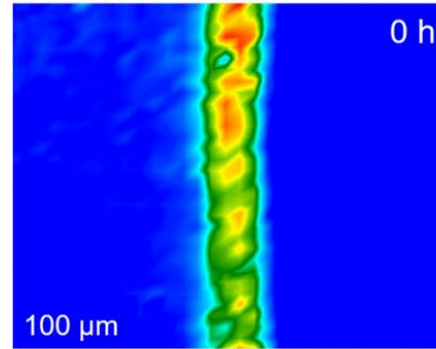


Fig.9 η of TPUs (with additive) [6]



03 Self-healing Efficiency

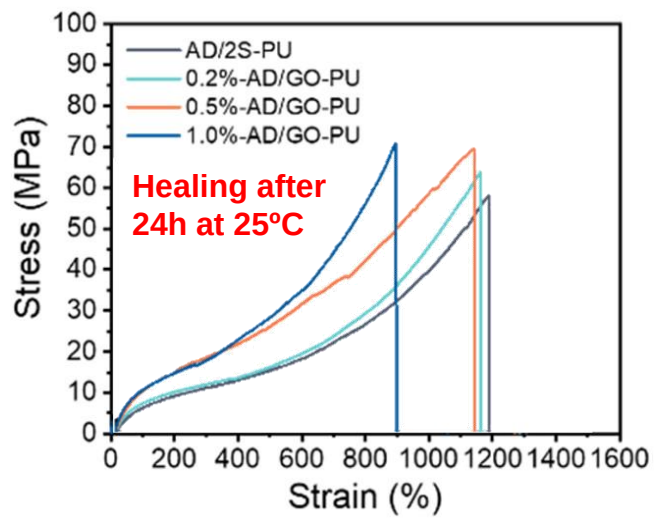


Fig.8 σ - ϵ curves of TPUs before and after healing(with additive) [6]

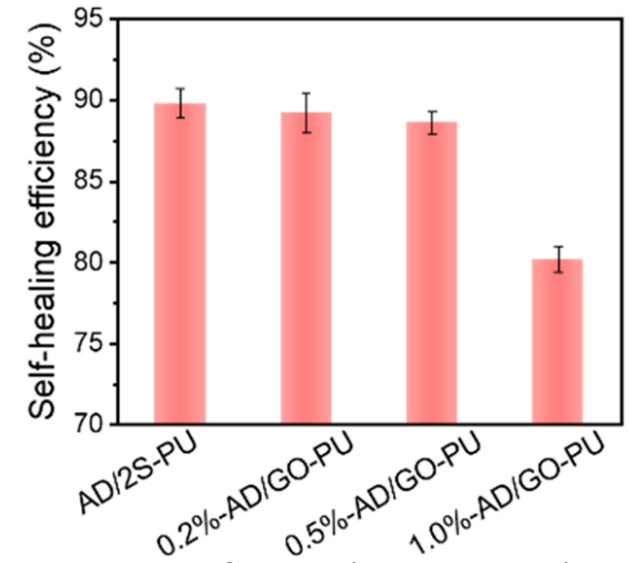


Fig.9 η of TPUs (with additive) [6]

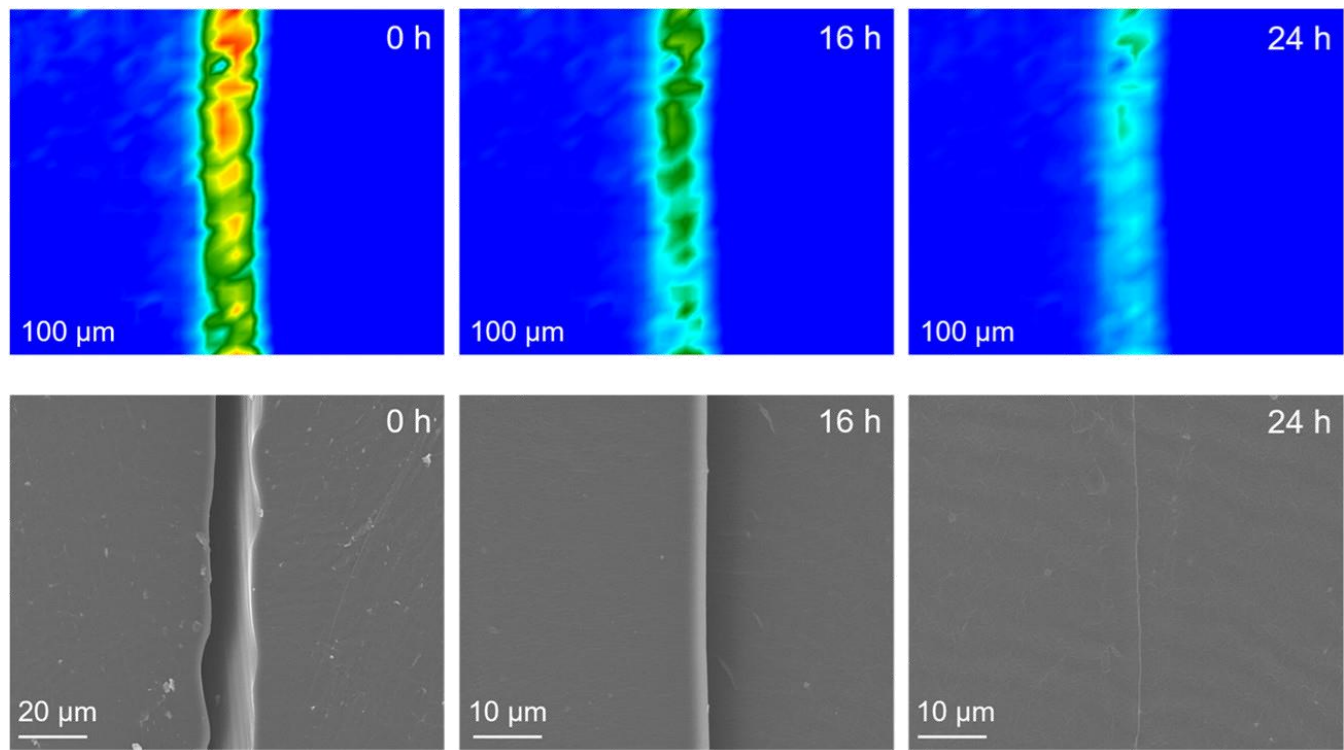


Fig.10 The infrared spectrum and SEM of 0.5%GO-AD/2S-PUT

	Original (MPa)	Restored (MPa)
Ultimate tensile strength	64.6	58.0
	78.3	69.4

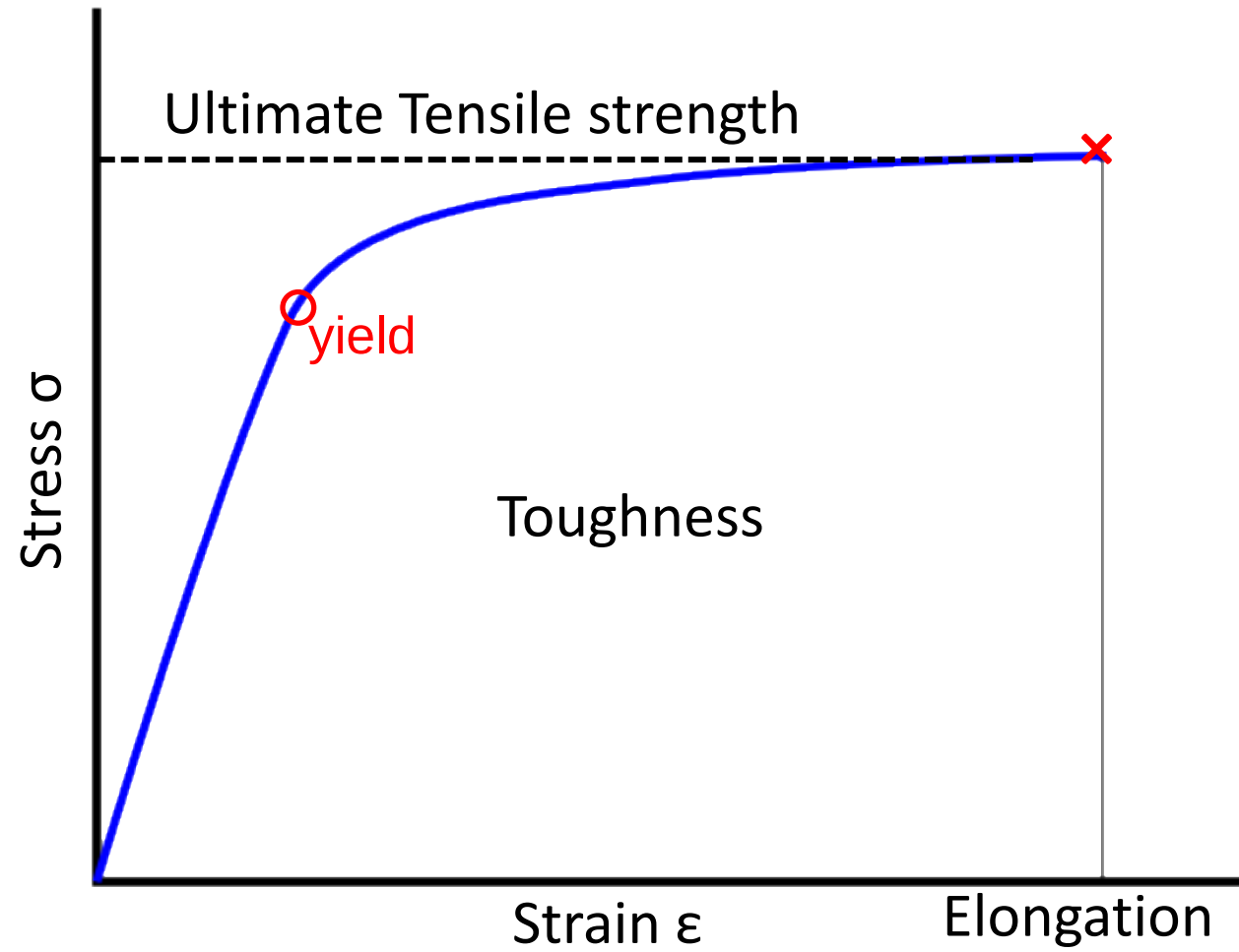
η reaches **89.8%~91.2%**

High Efficiency



03

Good Mechanical Performance





03

Good Mechanical Performance

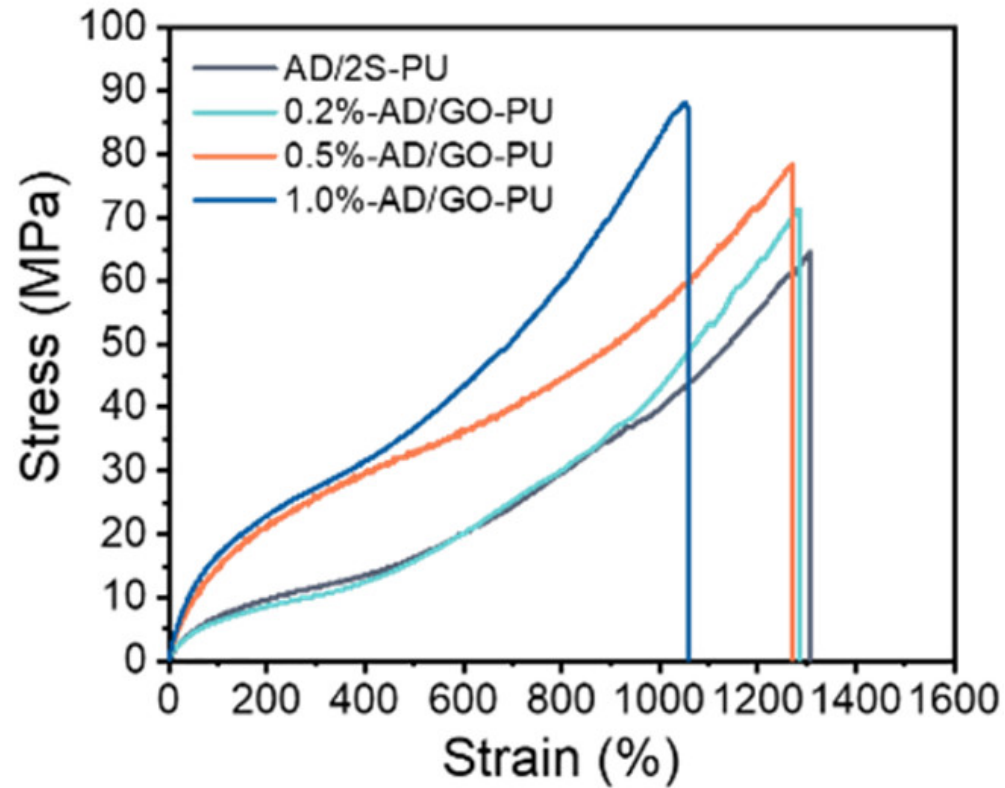


Fig.11 σ - ϵ curves of TPUs (with additive) [6]

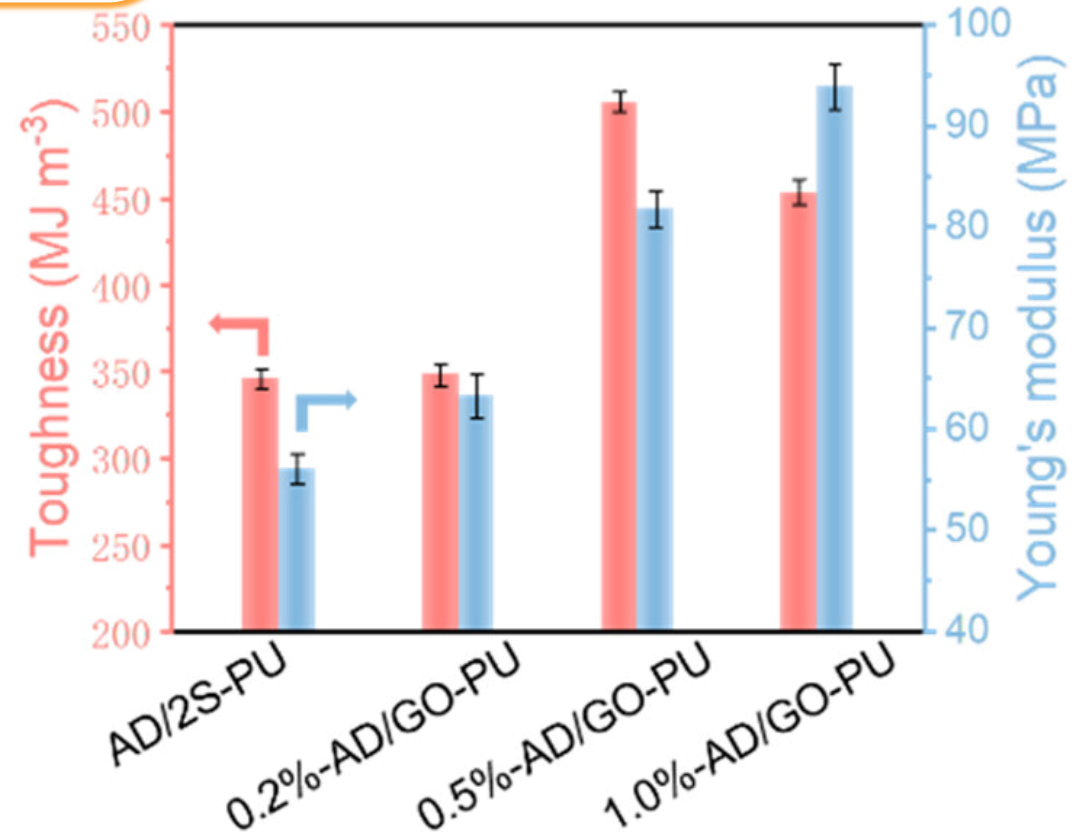


Fig.12 Toughness and moduli of TPUs (with additive) [6]

- UTS 64.6 MPa
- Toughness 345.8 MJ m⁻³
- Young's Modulus 56.1 MPa
- Elongation 1306.8 %

0.5%AD-GO

- UTS 78.3 MPa
- Toughness 505.7 MJ m⁻³
- Young's Modulus 81.8 MPa
- Elongation 1273.2 %



03

Good Mechanical Performance

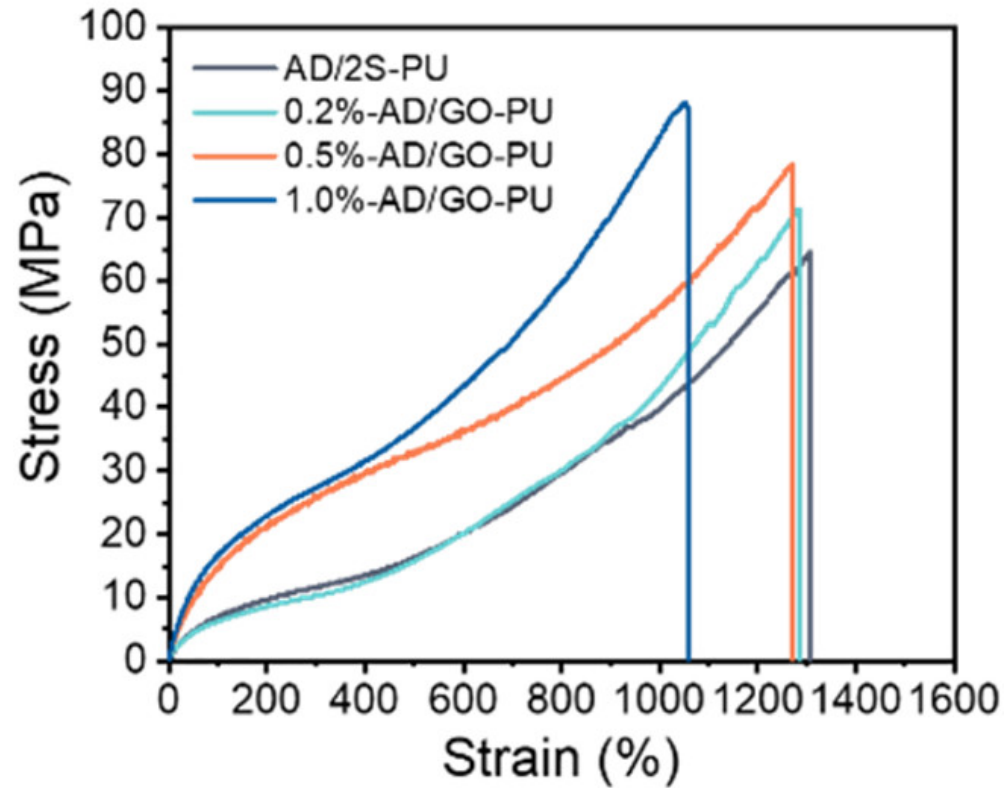


Fig.11 σ - ϵ curves of TPUs (with additive) [6]

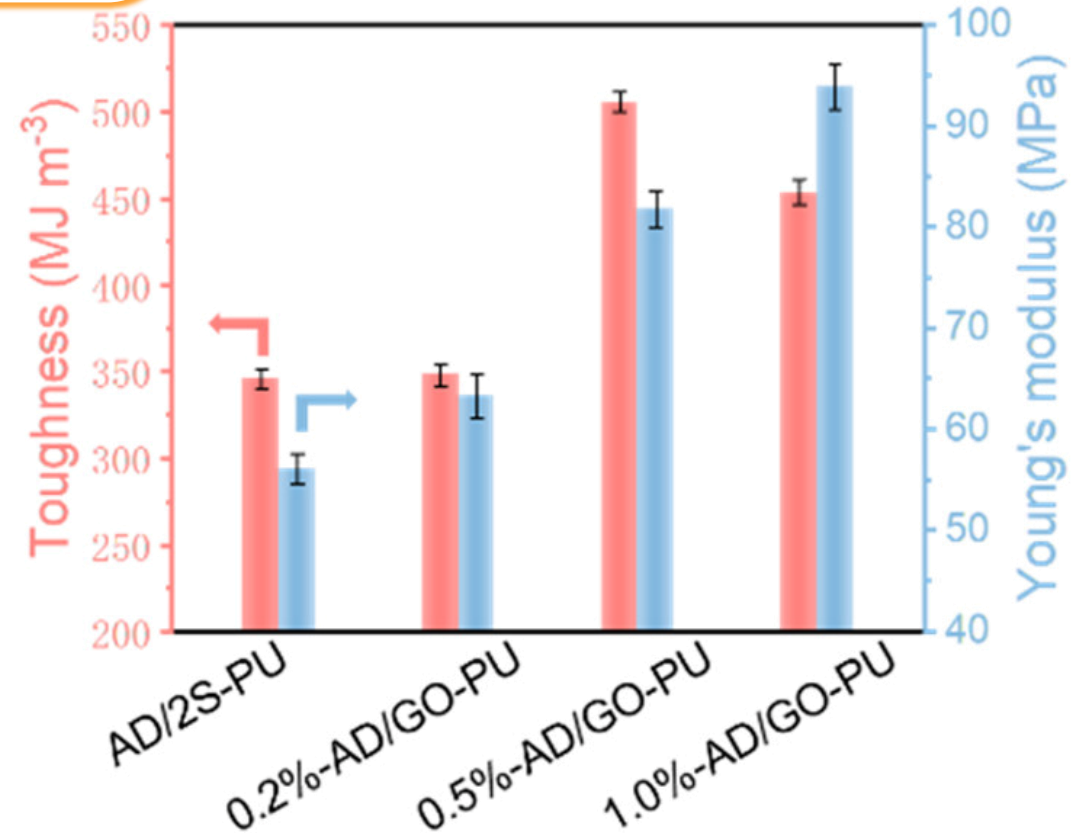


Fig.12 Toughness and moduli of TPUs (with additive) [6]

● UTS	64.6 MPa	High Mechanical Performance Highly Enforceable by Additive	● UTS	78.3 MPa
● Toughness	345.8 MJ m ⁻³		● Toughness	505.7 MJ m ⁻³
● Young's Modulus	56.1 MPa		● Young's Modulus	81.8 MPa
● Elongation	1306.8 %		● Elongation	1273.2 %

0.5%AD-GO

04



Application



04

Application of TPU

Good tensile ability

Rubber-like elasticity

Self-healing

Modification potential

Various
applications

Auto industry



Consumer electronics



Architecture



Healthcare





04

Paint Protection Film (PPF) for cars

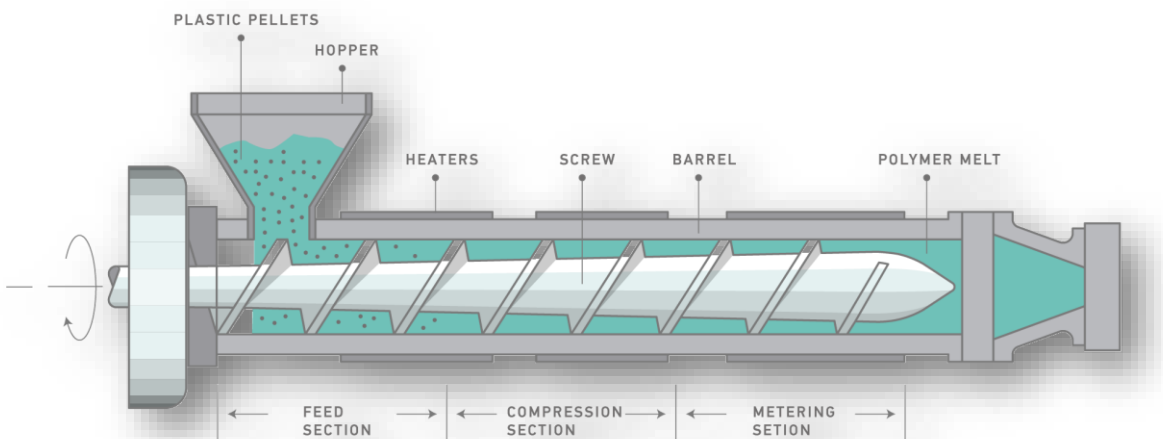


Fig.13 Schematic diagram of TPU processing method



Processing Method: **Extrusion** [7]

- TPU is formed into a film by **extrusion process**
- **Surface-treated** into a PPF
- preserve your vehicle's polished look

Easy to Customize : **Thermoplastic and toughness**

- have strong **plasticity** and **toughness** to fit different cars

Long-lasting Protection: **Self Healing and low T_g**

- Minor scratches can get **self-healed**
- T_g is low and the **service temperature range** is wide
- **Abrasion resistance** protect the paints from sand.



04

Artificial Blood Vessel [8]



Bionic Self-healing

- Able to **self-seal** around the needle hole
- Inner and outer surfaces shows **micro seam**, favorable for hemostasis
- **High mechanical properties** after multiple punctures

Flexibility

- Since TPU consists of soft and hard segments, its **low modulus** of elasticity and high toughness contribute to high flexibility

outer wall of artificial blood vessel using TPU

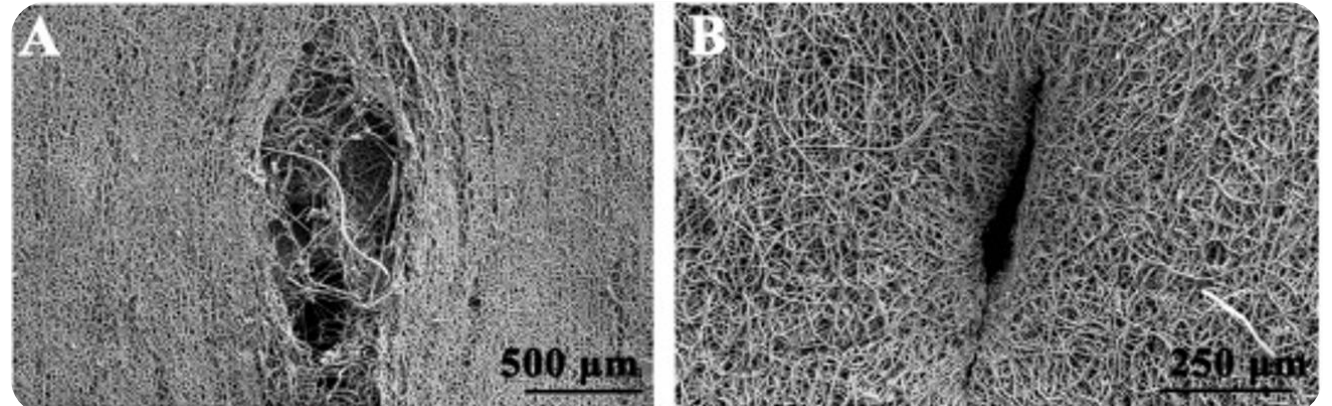


Fig.14 SEM morphology of inner (A) and outer (B) layers of TPU vessel wall after penetration

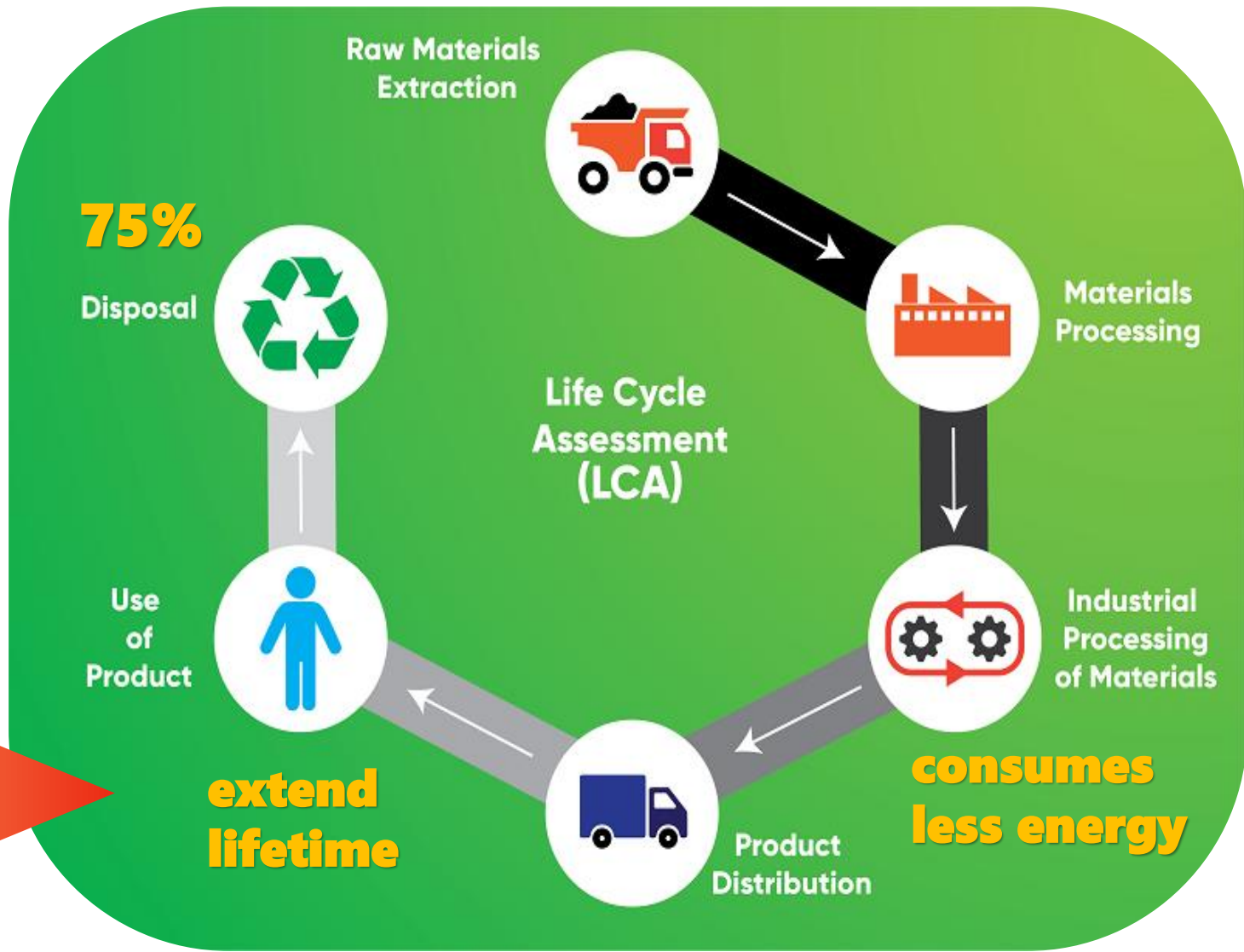
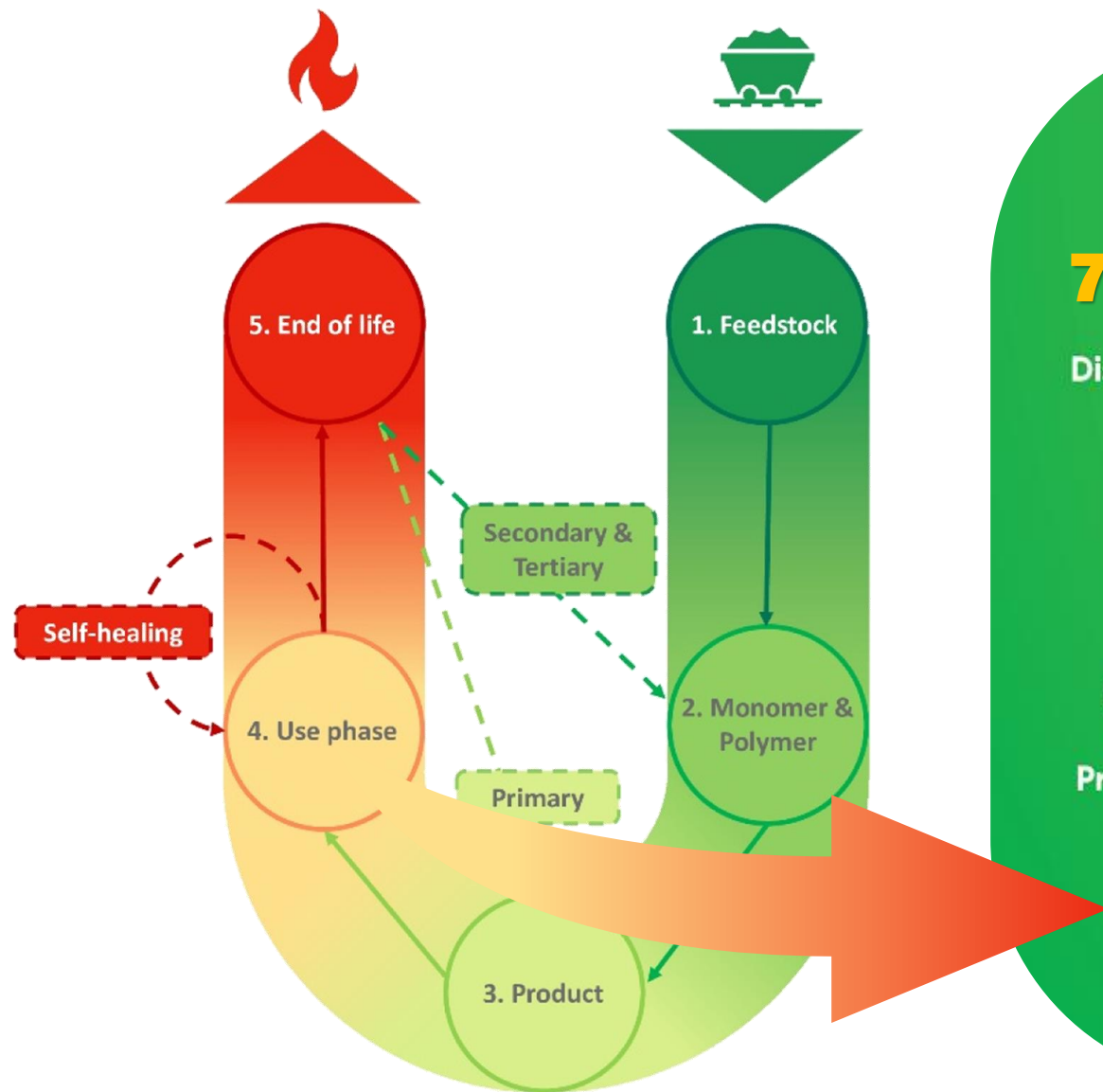
05



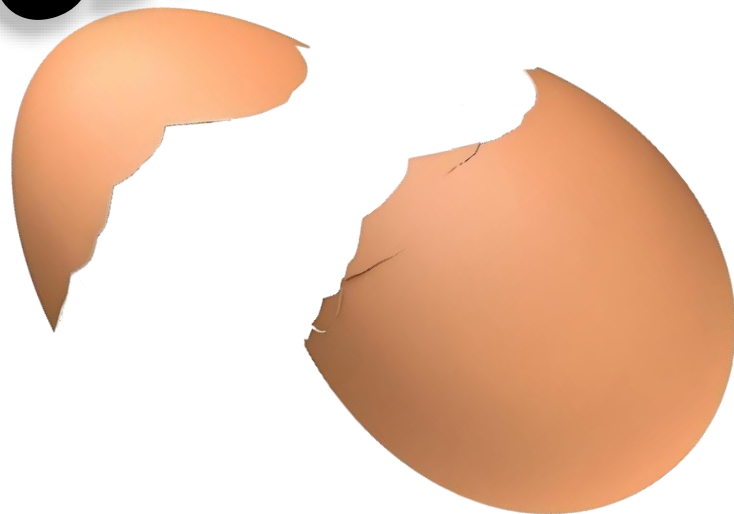
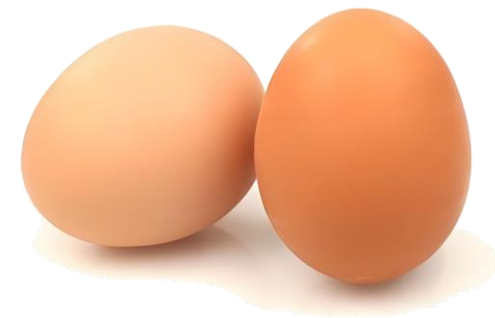
Sustainability



05 Sustainability



You **Can!**



unscramble an egg



Yuxin Liu



Jingyu Chen



Zishan Ding



Tianjian Feng



Zeying Jin



Guozhi Xu

References

- [1] Eun Joo Shin, Jae Hyun Son, Hyeri Jun & Sunhee Lee. (2024). Manufacturing of Bio-Based TPU/CNT Composites Using Solvent-Free In-Situ Polymerization for 3D Printing Filament Applications. *Fibers and Polymers* (prepublish), 1-9
- [2] Fuhao Yu, Xiaoli Yu, Pengfei Jia, Xiaodong Zhou, Hui Wang, Yanbei Hou... & Yuan Hu. (2024). Synergic effect between La-MOF and MXene: Revolutionizing TPU with improved mechanical strength and fire safety. *Chemical Engineering Journal* 156145-156145.
- [3] Dimitrios G. Papageorgiou (2024) Mechanical Properties of Polymers. *QMES Physical Properties of Polymers*
- [4] Zhu X, Zhang W, Lu G, Zhao H, Wang L. Ultrahigh Mechanical Strength and Robust Room-Temperature Self-Healing Properties of a Polyurethane–Graphene Oxide Network Resulting from Multiple Dynamic Bonds. *ACS Nano*. 2022 Oct 10;16(10):16724–35.
- [5] Li Y, Jin Y, Fan W, Zhou R. A review on room-temperature self-healing polyurethane: synthesis, self-healing mechanism and application. *Journal of Leather Science and Engineering*. 2022 Aug 30;4(1).
- [6] Zhu, X., Zhang, W., Lu, G., Zhao, H., & Wang, L. (2022). Ultrahigh Mechanical Strength and Robust Room-Temperature Self-Healing Properties of a Polyurethane–Graphene Oxide Network Resulting from Multiple Dynamic Bonds. *ACS Nano*, 16(10), 16724–16735. <https://doi.org/10.1021/acsnano.2c06264>.
- [7] *What is Extrusion? Plastic Extrusion Process*. (n.d.). Elastron Kimya. Retrieved November 11, 2024, from <https://www.elastron.com/en/what-is-extrusion-plastic-extrusion-process-elastron-tpe>
- [8] Fang, Z., Xiao, Y., Geng, X., Jia, L., Xing, Y., Ye, L., Gu, Y., Zhang, A., & Feng, Z. (2022). Fabrication of heparinized small diameter TPU/PCL bi-layered artificial blood vessels and in vivo assessment in a rabbit carotid artery replacement model. *Biomaterials Advances*, 133, 112628. <https://doi.org/10.1016/j.msec.2021.112628>
- [9] Arena, G.; Friedrich, K.; Acierno, D.; E. Padenko; Russo, P.; Filippone, G.; Wagner, J. Solid Particle Erosion and Viscoelastic Properties of Thermoplastic Polyurethanes. *eXPRESS Polymer Letters* 2015, 9 (3), 166–176. <https://doi.org/10.3144/expresspolymlett.2015.18>.
- [10] Fundamentals of Product Life Cycle GHG Accounting – Green Blogs. Crown-monkey.com. <https://greenblogs.crown-monkey.com/product-life-cycle-accounting-and-reporting-standard-fundamentals-of-product-life-cycle-ghg-accounting/> (accessed 2024-11-11).
- [11] Cerdan, K.; Thys, M.; Aleix Costa Cornellà; Demir, F.; Norvez, S.; Vendamme, R.; Peter Van Puyvelde; Joost Brancart. Sustainability of Self-Healing Polymers: A Holistic Perspective towards Circularity in Polymer Networks. *Progress in polymer science* 2024, 152, 101816–101816. <https://doi.org/10.1016/j.progpolymsci.2024.101816>.
- [12] Zhang, J.; Lv, S.; Zhao, X.; Sun, Y.; Ma, S.; Zhou, F. Biobased, Self-Healing, and Recyclable Polyurethane Derived Hydrogel-Elastomer Hybrids for Efficient Lubrication. *Progress in Organic Coatings* 2024, 188, 108212. <https://doi.org/10.1016/j.porgcoat.2024.108212>.